Financial and Operational Analysis of U.S. Hospitals Using Integrated CMS–BLS–Census Data and Machine Learning (2011–2022)

Interim Capstone Report

Course: Capstone Project 2
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Abstract

Hospital financial sustainability in the United States is shaped by a complex interaction of institutional efficiency, workforce labor costs, geographic accessibility, and community socioeconomic demand. Existing hospital financial studies frequently analyze these determinants independently, limiting the ability to understand combined drivers of hospital efficiency and financial risk.

This project develops a comprehensive, machine-learning-ready dataset integrating Centers for Medicare & Medicaid Services (CMS) hospital cost reports with Rural–Urban Continuum Codes (RUCC), Bureau of Labor Statistics (BLS) healthcare wage data, and U.S. Census socioeconomic indicators across the period 2011–2022. The integration enables joint analysis of institutional, geographic, workforce, and community determinants of hospital financial outcomes.

Using this dataset, predictive models were developed for hospital Profit Margin and Cost-to-Charge Ratio (CCR) employing baseline regression, tree-based ensemble, gradient boosting, and kernel methods. Random Forest achieved the strongest predictive performance ($R^2 = 0.517$ for Profit Margin; $R^2 = 0.576$ for CCR).

Results indicate that hospital profitability is primarily determined by internal operational efficiency, whereas cost efficiency reflects both institutional performance and regional socioeconomic context. The project demonstrates a scalable healthcare data engineering and machine learning framework suitable for financial risk assessment, health system analysis, and policy research.

1. Introduction

Hospital financial sustainability has become an increasingly critical concern in the United States healthcare system. Over the past decade, numerous hospitals—particularly in rural and socioeconomically disadvantaged regions—have experienced financial distress or closure. Rising workforce costs, heterogeneous patient demand, constrained reimbursement systems, and regional socioeconomic disparities contribute to substantial variability in hospital financial performance.

Hospital financial outcomes are determined by multiple interacting dimensions:

- internal operational efficiency (capacity utilization, staffing intensity, throughput)
- financial structure and cost intensity
- regional healthcare labor market conditions
- community socioeconomic demand

- geographic accessibility and rurality

Most existing hospital financial studies rely primarily on institutional accounting data derived from cost reports. Such approaches neglect external drivers such as regional wage pressures and population socioeconomic burden, which significantly influence hospital cost structures and service demand.

This study addresses this limitation by constructing a unified multi-source hospital dataset integrating institutional, workforce, geographic, and socioeconomic information across time and geography. Machine learning methods are applied to model hospital financial efficiency using this integrated dataset.

Unlike prior hospital financial analyses based solely on institutional data, this project incorporates contextual workforce and socioeconomic factors within a unified predictive framework. This integration enables more realistic modeling of hospital financial efficiency across regions and time.

2. Background and Related Work

Healthcare financial modeling has traditionally focused on institutional accounting metrics derived from hospital cost reports. While these data provide detailed operational and financial information, they fail to capture external environmental factors that influence hospital costs and demand.

Recent applied healthcare analytics research demonstrates that predictive performance improves when contextual datasets—such as labor market indicators and population characteristics—are integrated with institutional data. Multi-source feature engineering enables models to capture environmental and structural influences that are not observable from internal accounting variables alone.

Studies in applied data science capstone projects across domains such as transportation safety and consumer behavior similarly emphasize the importance of integrating heterogeneous data sources and employing ensemble machine learning algorithms to model nonlinear relationships. These studies follow a standardized analytical workflow consisting of:

1. multi-source data acquisition
2. preprocessing and integration
3. feature engineering
4. model development
5. evaluation and interpretation

The present project adopts this established methodology by integrating CMS hospital cost reports with geographic classification (RUCC), workforce wage data (BLS), and socioeconomic indicators (Census) to model hospital financial efficiency using multiple machine learning algorithms.

3. Project Objectives

The primary objective of this capstone is to construct and analyze an integrated healthcare dataset capable of predicting hospital financial efficiency across the United States.

Specific objectives include:

1. Designing a multi-source data engineering pipeline integrating CMS, RUCC, BLS, and Census datasets
2. Engineering interpretable financial, operational, workforce, and socioeconomic features at hospital-year level
3. Constructing machine-learning-ready predictors and targets for Profit Margin and CCR
4. Implementing baseline, ensemble, gradient boosting, and kernel-based regression models
5. Applying cross-validated preprocessing pipelines for mixed-type predictors
6. Evaluating model performance and selecting optimal algorithms
7. Interpreting drivers of hospital financial efficiency
8. Demonstrating a scalable healthcare analytics framework

4. Data Sources and Analytical Roles

Table 4.1. Data Sources and Analytical Roles

Source	Granularity	Analytical Role
CMS HCRIS	Hospital–Year	Institutional operations & finance
RUCC	County	Geographic rurality
BLS OEWS	State–Year–Occupation	Workforce labor cost
Census ACS	County–Year	Socioeconomic context

CMS data define observational units and target variables, while RUCC, BLS, and Census provide contextual predictors representing environmental influences on hospital performance.

5. CMS Hospital Data Engineering

CMS hospital cost reports provide detailed financial and operational data including revenue, expenses, charges, staffing, bed capacity, and utilization measures. However, raw CMS variables vary widely in scale and are strongly correlated with hospital size, limiting comparability across institutions.

To enable meaningful modeling, normalized performance indicators were engineered.

5.1 Financial Performance Indicators

$$Profit\ Margin\ =\ Net\ Income \div Total\ Revenue$$

$$CCR\ =\ Operating\ Costs \div Total\ Charges$$

These ratios measure profitability and cost efficiency independent of hospital scale.

5.2 Operational Efficiency Indicators

Operational performance variables were normalized by capacity:

- revenue per bed
- discharges per bed
- staff-to-bed ratio
- average length of stay

These features capture throughput and resource utilization.

5.3 Care Intensity Indicators

Healthcare service burden was represented by:

- charity care ratio
- uncompensated care ratio

These variables reflect socioeconomic demand pressures on hospitals.

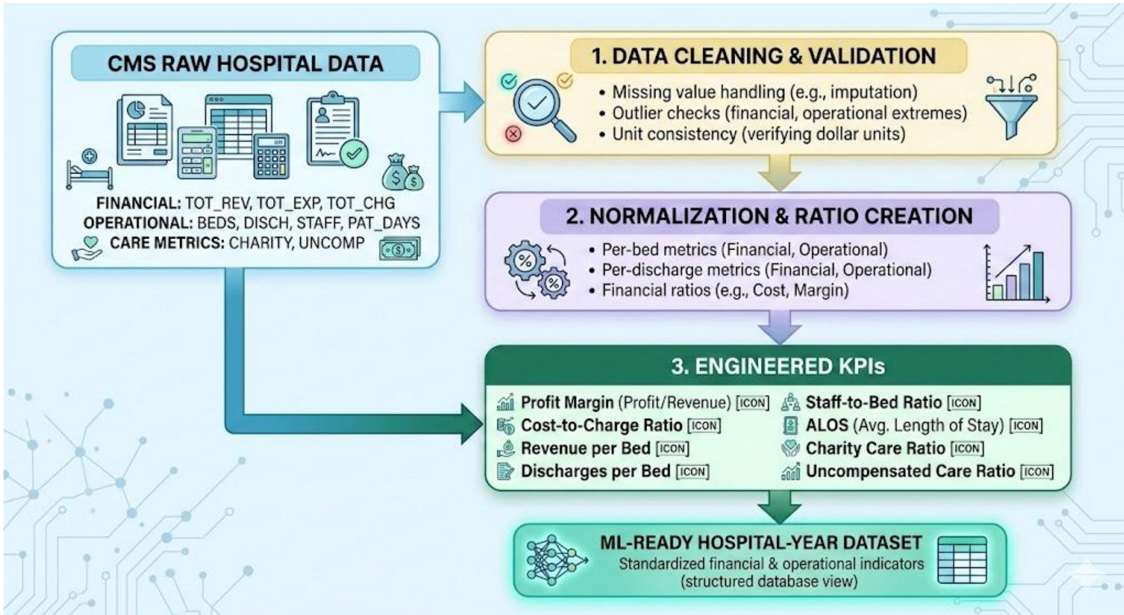

Figure 5.1. CMS KPI Feature Construction Pipeline.

Raw CMS hospital cost report variables are cleaned, normalized, and transformed into standardized financial, operational, and care-burden indicators at the hospital-year level for machine learning modeling.

6. RUCC Geographic Integration

Geographic accessibility significantly influences hospital demand patterns and financial viability. RUCC codes classify counties by population size and proximity to metropolitan areas.

CMS data do not include county FIPS identifiers; therefore, a county normalization pipeline was implemented including uppercase standardization, suffix removal, punctuation stripping, and special-case mapping.

RUCC values were transformed into ordinal rucc_code and binary rural_urban.

$$rural\_urban = Urban\ if\ rucc\_code \leq 3$$

$$rural\_urban = Rural\ if\ rucc\_code \geq 4$$

Observed missingness (~8–11%) corresponds to unmatched counties and special jurisdictions rather than merge errors.

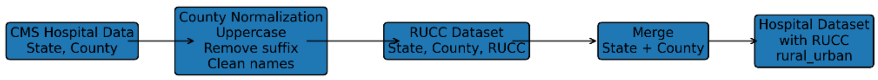

Figure 6.1. RUCC Merge and Normalization Process.
County names from CMS hospital data are standardized and matched to RUCC county classifications to assign rural–urban indicators.

7. BLS Workforce Wage Data Engineering

Labor costs represent the largest component of hospital operating expenses. Regional wage variation strongly influences hospital financial performance. BLS Occupational Employment and Wage Statistics datasets (2011–2022) were standardized to create a unified multi-year wage dataset.

Healthcare occupations were identified using SOC codes:

- 29-xxxx (healthcare practitioners)
- 31-xxxx (healthcare support)

Occupations were grouped into physician, practitioner, registered nurse, support, and other categories

7.1 Employment-Weighted Wage Calculation

$$W = \frac{\sum(a\_mean \cdot tot\_emp)}{\sum tot\_emp}$$

This weighting reflects the actual wage structure faced by hospitals.

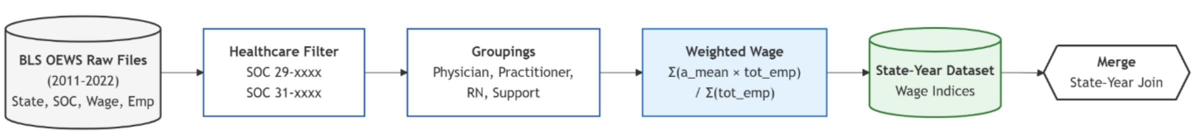

Figure 7.1. BLS Wage Aggregation Pipeline.
Raw occupational wage data are filtered to healthcare occupations, grouped by role category, and aggregated into employment-weighted state-year wage indices.

8. Census Socioeconomic Integration

Community socioeconomic conditions influence hospital service demand, charity care burden, and payer mix. County-level ACS 5-year estimates (2012–2022) were retrieved including population, median household income, poverty rate, education level, and median age. ACS data for 2011 are structurally unavailable and retained as expected missingness.

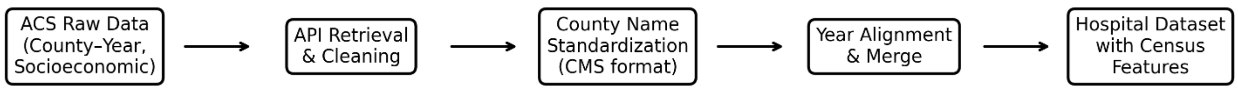

Figure 8.1. Census Data Processing Pipeline.
County-level ACS socioeconomic indicators are retrieved, standardized, and merged to hospital records by county-year.

9. Multi-Source Dataset Integration

A staged integration pipeline aligned all datasets to hospital-year level:

- Stage 1 — CMS + RUCC
- Stage 2 — CMS-RUCC + Census
- Stage 3 — Addition of BLS

Initial merging of occupation-level BLS data created a many-to-many expansion (>4 GB). Aggregating wages prior to merging preserved hospital-level granularity and manageable dataset size.

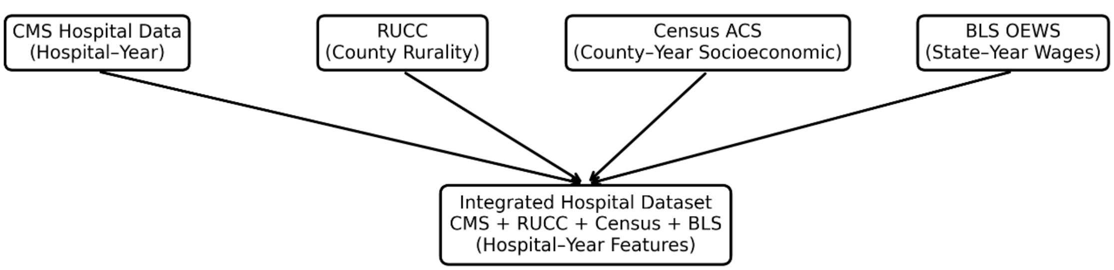

Figure 9.1. Integrated CMS–RUCC–Census–BLS Architecture

10. Final Integrated Dataset Characteristics

The final integrated dataset contains:

- 75,870 hospital-year observations
- 56 variables
- one row per hospital-year

Feature categories include financial ratios, operational metrics, workforce wages, socioeconomic indicators, and geographic classification.

Feature Category	***Example Variables***	***Data Source***	***Analytical Role***
Financial Performance	*profit_margin, ccr*	*CMS HCRIS*	*Target variables measuring hospital profitability and cost efficiency*
Operational Efficiency	*revenue_per_bed, discharges_per_bed, staff_to_bed, alos*	*CMS HCRIS*	*Internal hospital throughput and resource utilization*
Care Burden	*charity_ratio, uncomp_care_ratio*	*CMS HCRIS*	*Socioeconomic service demand on hospitals*
Geographic	*rucc_code, rural_urban*	*RUCC*	*Rurality and accessibility classification*
Workforce Wages	*wage_physician, wage_practitioner, wage_rn, wage_support*	*BLS OEWS*	*Regional healthcare labor cost environment*
Socioeconomic Context	*population, median_income, poverty_rate, education, median_age*	*Census ACS*	*Community demographic and economic conditions*

Table 10.1 Integrated Dataset Summary

Dataset Characteristics

- *Observations: 75,870 hospital–year records*
- *Predictors (ML): 44 features*
- *Total variables: 56*
- *Granularity: Hospital–Year*
- *Coverage: United States, 2011–2022*

11. Machine Learning Dataset Preparation

From the integrated dataset, an ML-ready matrix (~55,644 rows, 44 predictors) was constructed. Preparation steps included removal of leakage variables, handling structural missingness, categorical encoding, numerical scaling, and train-test split.

Figure 11.1. ML Dataset Preparation Pipeline

12. Machine Learning Methodology

12.1 Baseline Models

Linear Regression, K-Nearest Neighbors, and Decision Tree models were implemented as interpretable benchmarks illustrating limitations of linear and local methods.

12.2 Advanced Models and Preprocessing Pipelines

Advanced algorithms included Random Forest, XGBoost, LightGBM, Ridge Regression, and Support Vector Machine. Tree-based models were scale-invariant, whereas SVM and linear models required standardized numerical predictors.

12.3 Cross-Validation Strategy

Cross-validation was applied during advanced model training to obtain stable performance estimates and prevent preprocessing leakage.

13. Model Performance Results

Table 13.1 Model Performance Summary (Test R^2)

Model	Profit Margin	CCR
Linear Regression	0.189	0.102
KNN	0.370	0.136
Decision Tree	−0.070	0.153
Random Forest	**0.517**	**0.576**
XGBoost	0.493	0.495
LightGBM	—	0.479
Ridge	0.136	—

Tree-based ensemble models substantially outperform linear and local methods, indicating nonlinear interactions among predictors. CCR prediction achieved higher R^2 than Profit Margin, suggesting cost efficiency is more structurally determined by observable predictors.

14. Model Selection

Random Forest provided the strongest combination of predictive accuracy, stability, and interpretability across both targets and was selected as the final modeling approach.

15. Prediction and Explainability

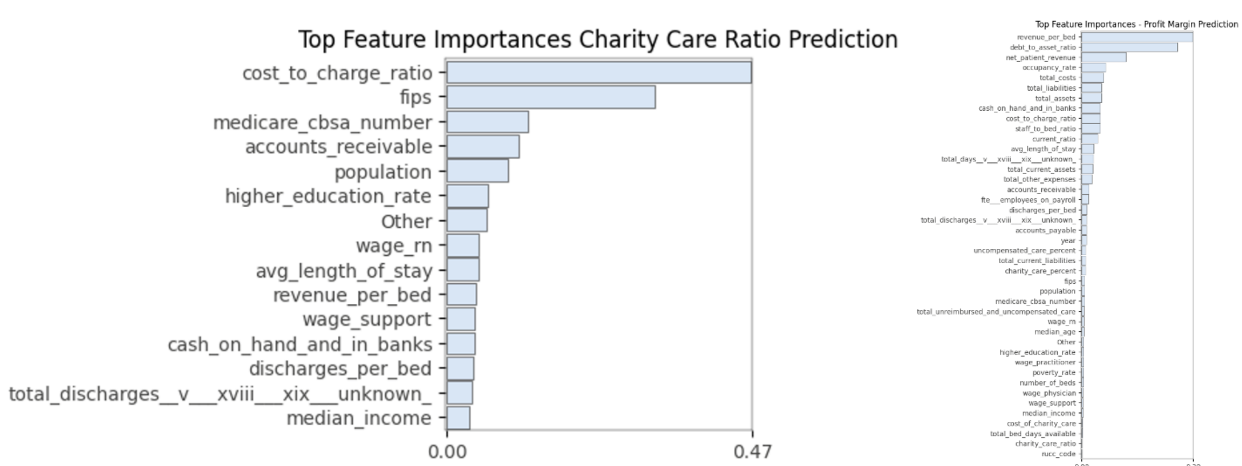

Figure 15.1. Random Forest Feature Importances for Profit Margin and Charity Care Ratio Prediction.

Workforce wages and socioeconomic context strongly influence CCR, while internal efficiency metrics dominate profitability prediction.

16. Key Findings

- Profitability depends primarily on internal operational efficiency
- Cost efficiency depends on institutional and community context
- Workforce wages significantly influence hospital costs
- Socioeconomic conditions affect CCR
- Ensemble ML best captures healthcare financial complexity

17. Discussion

The findings of this study show that hospital financial efficiency is shaped by both internal operational performance and external environmental conditions. Profitability is primarily associated with internal efficiency indicators such as revenue per bed, discharges per bed, and staffing intensity, suggesting that hospitals can influence margins through throughput and resource utilization. In contrast, cost efficiency measured by the Cost-to-Charge Ratio (CCR) reflects both institutional performance and regional context, including workforce wage levels and community socioeconomic conditions. This distinction indicates that while profitability is largely operationally determined, hospital cost structures are partly constrained by external labor and demand environments.

The superior performance of ensemble machine learning models—particularly Random Forest—compared with linear and kernel methods confirms the presence of nonlinear relationships among hospital financial determinants. Hospital performance arises from interactions between staffing, capacity, labor costs, and geographic accessibility that cannot be adequately captured by additive models. The results therefore support the suitability of tree-based ensemble learning for healthcare financial modeling, as these approaches effectively capture heterogeneous predictor influence across hospitals and regions.

A key contribution of this study is demonstrating the value of integrating contextual workforce and socioeconomic datasets with institutional hospital data. Traditional hospital financial analyses relying solely on cost reports may over-attribute financial variation to internal efficiency differences. The improved predictive accuracy observed after incorporating BLS wage and Census socioeconomic indicators—particularly for CCR—shows that hospitals operate within regionally constrained cost and demand environments. Overall, the integrated multi-source dataset and ensemble modeling framework provide a more realistic and comprehensive approach to analyzing hospital financial performance across geographic settings.

18. Conclusion

This study developed a comprehensive, machine-learning-ready healthcare analytics dataset integrating institutional hospital cost report data (CMS HCRIS) with geographic rurality classifications (RUCC), regional healthcare labor market indicators (BLS OEWS), and county-level socioeconomic characteristics (U.S. Census ACS) across the period 2011–2022. The integration of these heterogeneous sources enabled simultaneous modeling of internal hospital efficiency and external environmental determinants of financial performance, addressing limitations of prior hospital financial analyses that rely solely on institutional accounting data.

Using this integrated dataset, multiple baseline and advanced machine learning models were implemented to predict two key hospital financial efficiency indicators: Profit Margin and Cost-to-Charge Ratio (CCR). Results demonstrate that ensemble machine learning models particularly Random Forest effectively capture the complex nonlinear relationships between operational efficiency, workforce costs, geographic accessibility, and socioeconomic context that shape hospital financial outcomes. Predictive performance was consistently higher for CCR than for Profit Margin, indicating that cost efficiency is more strongly determined by observable structural and environmental factors, whereas profitability is influenced to a greater extent by internal managerial and financial dynamics not fully captured in available data.

Feature importance analysis further revealed that hospital operational throughput indicators (e.g., revenue per bed, discharges per bed, staffing intensity) are dominant predictors of profitability, while regional wage levels and community socioeconomic conditions exert substantial influence on cost efficiency. These findings highlight the dual nature of hospital financial sustainability: profitability depends primarily on internal efficiency and capacity utilization, whereas cost efficiency reflects both institutional performance and external demand and labor market pressures.

Beyond predictive modeling results, the project contributes a scalable multi-source healthcare data engineering framework that aligns institutional, geographic, workforce, and socioeconomic data at the hospital-year level. This integrated dataset and modeling pipeline provide a reproducible foundation for future research in hospital financial risk assessment, rural health policy analysis, workforce planning, and health system performance evaluation. The methodology demonstrates how contextual socioeconomic and labor market data can substantially enhance predictive healthcare analytics compared with institution-only approaches.

Overall, this study demonstrates the feasibility and analytical value of integrating heterogeneous national datasets to model hospital financial efficiency at scale. Ensemble machine learning models, particularly Random Forest, provide robust predictive capability and interpretable insights into the structural drivers of hospital cost efficiency and profitability across the United States healthcare system. The integrated data engineering and machine learning framework developed in this project offers a practical and extensible approach for healthcare analytics and policy-relevant financial modeling.

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Appendix

Appendix A — CMS Feature Engineering Implementation

This appendix documents the transformation of raw Centers for Medicare & Medicaid Services (CMS) Hospital Cost Report Information System (HCRIS) variables into normalized financial and operational performance indicators used in predictive modeling.

A.1 Source Data Structure

CMS hospital cost report data were extracted at the hospital-year level. The primary financial and operational variables used in feature engineering are listed in Table A1.

Table A1. Raw CMS Variables Used in Feature Engineering

Variable	***Description***	***Units***
TOT_REV	Total hospital revenue	USD
TOT_EXP	Total hospital expenses	USD
TOT_CHG	Total hospital charges	USD
BEDS	Staffed beds	count
DISCH	Discharges	count
PAT_DAYS	Patient days	count
STAFF	Total staffing	FTE
CHARITY	Charity care	USD
UNCOMP	Uncompensated care	USD

These raw variables are highly correlated with hospital size and case mix. Direct use would bias machine learning models toward larger hospitals; therefore, normalization was required.

A.2 Financial Outcome Variables

Two primary financial outcomes were engineered.

Profit Margin

$$ProfitMargin = \frac{TOT\_REV - TOT\_EXP}{TOT\_REV}$$

Interpretation: proportion of revenue retained after expenses.

Cost-to-Charge Ratio (CCR)

$$CCR = \frac{TOT\_EXP}{TOT\_CHG}$$

Interpretation: cost efficiency relative to billed charges.

A.3 Operational Efficiency Indicators

Operational metrics were normalized by capacity or throughput to allow cross-hospital comparability.

Revenue per Bed

$$RevenuePerBed = \frac{TOT\_REV}{BEDS}$$

Measures revenue productivity per capacity unit.

Discharges per Bed

$$DischargesPerBed = \frac{DISCH}{BEDS}$$

Measures bed turnover and utilization.

Staff-to-Bed Ratio

$$StaffToBed = \frac{STAFF}{BEDS}$$

Measures staffing intensity.

Average Length of Stay (ALOS)

$$ALOS = \frac{PAT\_DAYS}{DISCH}$$

Measures care duration and case complexity.

A.4 Care Burden Indicators

Community service burden was represented by normalized care ratios.

Charity Care Ratio

$$CharityRatio = \frac{CHARITY}{TOT\_REV}$$

Uncompensated Care Ratio

$$UncompCareRatio = \frac{UNCOMP}{TOT\_REV}$$

These variables capture socioeconomic demand pressures on hospitals.

A.5 Implementation Details

Feature engineering was implemented in the following notebooks:

- CMS_Urban_Rural_Dataset.ipynb
- CMS_Urban_Rural_BLS_Dataset.ipynb
- ML_Dataset_Creation.ipynb

Derived variables were stored in:

- hospital_kpi_ready_with_rucc.csv

Appendix B — RUCC Geographic Merge Procedure

This appendix describes the geographic linkage of CMS hospital records to Rural–Urban Continuum Codes (RUCC).

B.1 Challenge

CMS HCRIS data contain county names but not standardized FIPS codes. RUCC datasets use county identifiers; therefore name-based normalization was required.

B.2 County Name Normalization Algorithm

CMS county names were standardized using the procedure in Table B1.

Table B1. County Normalization Steps

Step	***Example***
Convert to uppercase	“Cook” → “COOK”
Remove suffixes	“COOK COUNTY” → “COOK”
Remove punctuation	“ST. LOUIS” → “ST LOUIS”
Collapse spaces	“LA SALLE” → “LA SALLE”
Manual mapping	“LASALLE” ↔ “LA SALLE”

B.3 Merge Keys

CMS dataset:
STATE + COUNTY_NORMALIZED

RUCC dataset:
STATE + COUNTY

Merge type: left join (hospital-preserving)

B.4 Output Variables

Variable	***Description***
rucc_code	ordinal rurality
rural_urban	binary classification

Binary classification rule:

$$rural\_urban = Urban\ if\ rucc\_code \leq 3$$

$$rural\_urban = Rural\ if\ rucc\_code \geq 4$$

B.5 Merge Quality Assessment

- Match rate: ~89–92%
- Missing RUCC: ~8–11%
- Causes: unmatched counties, territories, naming discrepancies

B.6 Implementation Reference

Implemented in:

- CMS_Urban_Rural_Dataset.ipynb
- CMS_Urban_Rural_BLS_Dataset.ipynb

Output stored in:

- hospital_cms_census_bls_rucc.csv

Appendix C — BLS Wage Aggregation Methodology

This appendix documents the construction of regional healthcare labor cost indicators from Bureau of Labor Statistics Occupational Employment and Wage Statistics (OEWS) data.

C.1 Source Datasets

Multi-year BLS files were standardized and combined:

File
bls_all_years_combined.csv
bls_all_occupations_standardized.csv
bls_healthcare_grouped.csv
bls_hospital_wage_group_dataset.csv

C.2 Healthcare Occupation Identification

SOC codes were filtered:

- 29-xxxx → practitioners
- 31-xxxx → support

Grouped into categories (Table C1).

Table C1. Healthcare Wage Categories

Category	***SOC***
Physician	29-1
Practitioner	29-2
Registered Nurse	29-1141
Support	31
Other	remaining

C.3 Employment-Weighted Wage Calculation

Regional wages were calculated per state-year:

$$W_{state,year} = \frac{\sum(a\_mean \cdot tot\_emp)}{\sum tot\_emp}$$

Where:

- a_mean= mean wage
- tot_emp= employment count

This weighting reflects workforce composition.

C.4 ML Wage Features

Feature
wage_physician
wage_practitioner
wage_rn
wage_support

C.5 Implementation Reference

Implemented in:

- BLS_Healthcare_Data.ipynb
- Aggregated_BLS.ipynb
- combined_bls_files.ipynb

Appendix D — Census Data Retrieval and Processing

This appendix describes the integration of county-level socioeconomic variables from the U.S. Census American Community Survey (ACS).

D.1 Source Datasets

File
census_raw_2012_2022.csv
census_county_renamed_2012_2022.csv
census_ml.csv

D.2 Variables Extracted

Variable	***Meaning***
population	county population
median_income	household income
poverty_rate	% below poverty
education	% college
median_age	demographic

D.3 Processing Steps

1. ACS API retrieval
2. County renaming to CMS format
3. Year alignment
4. Merge to hospital dataset

D.4 Structural Missingness

ACS begins 2012 → 2011 missing by design.

D.5 Implementation Reference

Implemented in:

- Census_data.ipynb
- CMS_BLS_Census.ipynb

Appendix E — Machine Learning Dataset Construction

This appendix documents creation of the final modeling dataset.

E.1 Source Tables

Dataset
hospital_cms_census_bls_rucc.csv
hospital_kpi_ready_with_rucc.csv

E.2 Target Variables

Target
profit_margin
ccr

E.3 Predictor Selection

Excluded:

- variables used in targets
- raw financial components

Final predictors: 44 features.

E.4 Encoding Strategy

Categorical variables:

Variable
ownership
hospital_type
rural_urban

Encoding: One-hot encoding for nominal variables and ordinal encoding for ordered categorical variables.

E.5 Scaling Strategy

Standardization applied to:

- SVM
- Ridge
- Linear models

Scaler: StandardScaler.

E.6 Dataset Split

Train/test split: random stratified.

Appendix F — Model Training and Evaluation

This appendix documents machine learning implementation.

F.1 Baseline Models

Model	***Notebook***
Linear Regression	Baseline_ML_Model.ipynb
KNN	Baseline_ML_Model.ipynb
Decision Tree	Baseline_ML_Model.ipynb

F.2 Advanced Models

Model	*Notebook*
Random Forest	Advanced_ML_Model.ipynb
XGBoost	Advanced_ML_Model.ipynb
LightGBM	Advanced_ML_Model.ipynb
SVM	Extended_ML_Model.ipynb
Ridge	Extended_ML_Model.ipynb

F.3 Cross-Validation

k-fold cross-validation applied during advanced training.

Appendix G — Model Artifacts

Trained models stored as .pkl files:

File
random_forest_profit_margin_model.pkl
random_forest_charity_care_ratio_model.pkl
linear_regression_profit_margin_model.pkl
k_nearest_neighbors_profit_margin_model.pkl
decision_tree_profit_margin_model.pkl

Artifacts located in:

ML Modelling With Models/

Appendix H — Analytical Notebooks

Complete workflow notebooks:

Notebook	***Purpose***
ML_Dataset_Creation.ipynb	final dataset
CMS_BLS_Census.ipynb	integration
Baseline_ML_Model.ipynb	baseline
Advanced_ML_Model.ipynb	ensemble
Extended_ML_Model.ipynb	advanced